Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 260 (2025) 626–634

Procedia Computer Science

www.elsevier.com/locate/procedia

Seventh International Conference on Recent Trends in Image Processing and Pattern Recognition (RTIP2R-2024)

Machine Learning and Time Series Approaches for Stock Price Prediction: A Case Study of Indian Banks

Tiya Chaudhary[a,*], Minirani S[b]

[a]Nilkamal School of Mathematics, Applied Statistics and Analytics (NSOMASA), NMIMS University, Mumbai - 400056
[b]Department of Mathematics, Mukesh Patel School Of Technology, Management and Engineering, NMIMS University, Mumbai-400056

Abstract

The Finance sector plays an important role in the economy of any country. With increased recognition of the stock market and its importance in the economy, correct investments and portfolio management has been the top priority in this sector. With the introduction of machine learning and time series models this has been easier. This research paper focuses on the comparative analysis of three most used machine learning and time series models namely Autoregressive Integrated Moving Average (ARIMA), Seasonal Autoregressive Integrated Moving Average (SARIMA) and Long Term Short Term Memory (LSTM) in predicting the stock prices of the Indian Banks namely ICICI Bank, IndusInd Bank, State Bank Of India, Kotak Mahindra Bank, HDFC Bank and Axis Bank using efficient and time saving means. Each models' accuracy is evaluated using error metrics like Root Mean Square Error, Mean Squared Error and Mean Absolute Error. The findings aim to provide an insight about the most preferred model in this field and also help financial analysts and investors to make the correct decisions while investing into the market.

© 2025 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the Seventh International Conference on Recent Trends in Image Processing and Pattern Recognition.

Keywords: ARIMA; SARIMA; LSTM; Root Mean Square Error; Mean Absolute Error; Stock price prediction.

1. Introduction

In the Indian banking sector, accurate forecasting of stock prices is a crucial task for financial analysts and investors. Financial markets are difficult to understand because they are very volatile and can be influenced by many factors. This paper examines the use of time series analysis and machine learning techniques like ARIMA, SARIMA, and LSTM in predicting stock prices for major Indian banks. These models aid in the exploitation of both linear and nonlinear elements in financial time series data, hence helping in understanding market movements. This research is, in essence, important in that it will help improve the investment decisions of both investors and finance professionals.

* Corresponding author.
E-mail address: chaudharytiya45@gmail.com

1877-0509 © 2025 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the Seventh International Conference on Recent Trends in Image Processing and Pattern Recognition.
10.1016/j.procs.2025.03.241

These models shall not only help in mitigating risks but also make optimal investments with funds, hence playing a role in better-informed financial planning and portfolio management. The other beneficiaries include individual/ institutional investors, financial advisors, portfolio managers who depend on accurate market forecasts to make decisions. The results of the study seek to shed some light on which forecasting approaches, traditional or modern ones, work better in predicting financial ratios in an Indian banking context and how they can provide potential future use cases for practicing accurate financial forecasts. The study uses the three principal models—ARIMA, SARIMA and LSTM to forecast stock prices from noted Indian banks. ARIMA (AutoRegressive Integrated Moving Average) is a widely used statistical analysis model that uses past data points to predict future trends. ARIMA uses the concepts of Autoregression, Moving Average and Differencing for non-stationary data. It has been well established that the model works in financial forecasting, with particular acumen for short-term predictions. But some of the limitations include difficulty in capturing complex patterns and seasonality that is present within data. The solution is to employ the SARIMA (Seasonal AutoRegressive Integrated Moving Average) model. ARIMA(Seasonal) or SARIMA that does take account of seasonality in our data if any. For handling seasonality, SARIMA (Seasonal AutoRegressive Integrated Moving-Average) model is employed. SARIMA is an extension of ARIMA which includes seasonal terms to deal with periodic patterns in data. This type of model is beneficial for markets like the financial market where seasonal effects play an important role. Previous studies have shown SARIMA can be more accurate than ARIMA for capturing the cyclical patterns present in financial time series data but still require past data and are computationally intensive to tune seasonal parameters. Hence, one can resort to an LSTM network, a kind of RNN. Since conventional mathematical models can be limited for these very reasons, LSTMs will be more appropriate for time series forecasting where the effect of past events on future results is very high in most cases. Contrasted with ARIMA and SARIMA, these are linear methods dealing with nonlinear relationships. In contrast, LSTM networks are capable of modeling complex relationships, hence better at understanding the data. While ARIMA and SARIMA are linear approaches, LSTM networks model complex, nonlinear relationships and give further nuanced insight into records. This capability makes LSTM especially powerful in handling complex realities of the stock market, many of which interrelate in nonlinear ways. However, LSTM requires large classroom realities and computational resources, which can affect heart overload, and also requires careful model tuning and validation.

2. Related Work

Studies have demonstrated the efficient use of traditional time series models such as ARIMA and SARIMA for short term stock price forecasting. For instance , J Patel et. al (2015) [1] applied the ARIMA model to forecast the Bombay Stock Exchange indices showcasing its potential in capturing stock price movements accurately. In Addition to this, in recent years advanced neural network models particularly Long Term Short Term Memory (LSTM) networks have shown promise in capturing long term dependencies and trends in stock prices outperforming traditional methods in some cases [2]. S Rastogi (2019) [3] discussed comparing several machine learning algorithms like Random Forest, Support Vector Machines and LSTM for stock market prediction using historical stock prices. The study concluded that LSTM provided the most accurate predictions due to its ability to capture temporal dependencies. Furthermore, H.T Leung (2020) [4] talked about comparing deep learning models like CNN and LSTM for the purpose of stock price prediction and it was concluded that LSTM model outperformed CNN due to its ability to handle sequential data effectively. In addition to this, an ensemble approach combining multiple machine learning algorithms was tested by Y.Shen(2018) [5] which showed that the ensemble approach improved prediction accuracy by leveraging the strengths of individual models. J.L Jones and C. Brown(2019) [6] conducted a study where they applied Holt Winters and SARIMA models to predict stock prices where they concluded that SARIMA performed better in capturing seasonal patterns in stock prices. Another research done in this field by E.Z.Zhang et.al (2019) [7] where they proposed a combined ARIMA and neural network hybrid approach model concluding that this approach improved prediction accuracy by capturing both linear and non linear components of the data. In addition to this, Uppala Meena Sirisha et. al(2022) [8] proposed a comparison between ARIMA, SARIMA and LSTM for a profit prediction analysis in this sector and concluded that LSTM turns out to outmodel ARIMA and SARIMA with an accuracy of 97.01% as stated in their paper. Further work done in this field is done by Aproova Mahadik et.al (2021) [9] where they concluded that ARIMA and LSTM are good models for stock traders to analyse and predict stock price data although ARIMA requires more time for processing compared to LSTM and LSTM performs well with large historical data and fewer

Date	Open	High	Low	Close	Adj Close	Volume
2002-07-01	18.500000	18.900000	18.299999	18.600000	13.320387	84409
2002-07-02	18.600000	19.000000	18.500000	18.650000	13.356194	34616
2002-07-03	18.650000	19.000000	17.650000	17.950001	12.854889	89445
2002-07-04	18.250000	18.350000	17.900000	18.000000	12.890697	58765
2002-07-05	17.850000	18.200001	17.650000	18.100000	12.962312	30847
...	...	...	...	...	...	...
2021-04-23	830.000000	859.950012	822.799988	850.000000	820.848450	7695324
2021-04-26	857.000000	870.950012	851.450012	859.549988	830.070984	4894557
2021-04-27	859.549988	886.000000	859.000000	882.450012	852.185608	5202133
2021-04-28	890.000000	931.000000	888.000000	926.900024	895.111084	9935828
2021-04-29	936.849976	958.849976	917.099976	939.450012	907.230713	12187280

4679 rows × 6 columns

Fig. 1: IndusInd Bank dataset

nan values. Omkar Deorukhar et. al (2019) [10] where they predicted next day closing prices using a comparison of sentiment Analysis, LSTM and ARIMA where they concluded that LSTM provided better accuracy and results compared to the other models. Sidra Mehtab et. al (2021) [11] predicting NIFTY 50 open values using four different deep learning models, one of them being LSTM which resulted in LSTM being the most appropriate model using one week prior data as mentioned by them. [12] talks about using 11 companies belonging to saudi stock exchange and implementing hyperparameter tuning which resulted in support vector regression to outperform other models used. [13] talks about using Holt winters exponential smoothing, ARIMA, Random Forest, MARS, simple RNN and LSTM on Infosys, ICICI bank and SUN pharmaceutical stocks over a period of January 2004 to December 2019 out of which LSTM turned out to be the best deep learning model for stock price prediction.

3. Methodology

The Stock price dataset has been obtained using yahoo finance website, and consists of approximately 4000 to 5000 rows depending on each bank ranging over a period of 20 years from 2001 to 2021. Figure 1 shows the dataset of one bank, IndusInd Bank for reference. The dataset contains columns namely the date of the stock price, the opening stock price, the highest stock price during that day, the lowest stock price during that day, the closing price for that day, the average closing price for that day and the volume of the stock traded for that day. The first step involved is data collection followed by data preprocessing, building the models and evaluating them. The collected data might contain some missing or NAN values due to no market trading days during public holidays and weekends. An important step is data preprocessing where handling these nan or missing values holds great importance for building a model with good accuracy and precise predictions. For accurate comparative analysis of these models the closing price of the stocks have been considered since the closing price gives a clear picture of the market at the end of each day.

3.1. Data Preprocessing

The chosen dataset is preprocessed by eliminating the nan or the missing values and grouping the data. The next step involves plotting the closing price of the stocks respectively to analyze the trend in the stocks. It was concluded that the stock prices had a steady increase from the year 2001 to 2008. A slight decrease has been seen from the year 2008 to 2010 due to the worldwide financial crisis which caused the stock market to crash. The stock price again caught a good momentum around the year 2012 until the year 2020. During the year 2020-2021 a decrease had again been seen due to the impact of COVID 19. After successfully analyzing the closing stock prices the next step is to build the models namely ARIMA, SARIMA and LSTM respectively.

3.2. Working of the ARIMA Model

The ARIMA model is the most frequently used model in predicting the future prices in the stock market. The model's main working principle is creating a link between the closing stock prices (in reference to this context) and the time. This relationship is further used to predict the future values. ARIMA model works best for patterned data which needs to be made stationary using certain tests and methods as mentioned in the building ARIMA model section. The ARIMA model is made up of different components with suitable roles namely AutoRegressive(AR), Integrated(I) and Moving Average (MA).[14]

The AutoRegressive component works on the principle of finding a relationship between the current and the past stock prices. The integrated component works on the principle of making the data stationary using methods of differencing to improve accuracy. The moving average component works on the principle of removing minor errors required for the model to run efficiently and provide stable results. The detailed working of the ARIMA model is described in the next section.

3.3. Building the ARIMA Model

ARIMA models work best with stationary time series datasets. The first step in building the ARIMA model involves checking if the data used is stationary (the statistical properties like mean, variance and standard deviation) are constant. This can be done using the ADF test and carrying out differencing. The ADF test works on the principle of null and alternative hypothesis where the null hypothesis states that if the test statistic is more negative than critical value and p value is less than 0.05 (significance level) then we reject the null hypothesis and the alternative hypothesis states that if the test statistic is less negative or greater than the critical value and p value is greater than the significance level then we accept the alternative hypothesis and conclude data is non stationary. Further processing includes differencing the data to make it stationary. Differencing is the technique of subtracting consecutive observations to eliminate trend and seasonality from the data to make it stationary. The number of differencing done to the data is the value of the ARIMA model parameter 'd'. Later the values for the model parameters namely p,d and q can be obtained. The traditional methods include plotting Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots. The ACF plot shows how well the present value of the series is related to its past values and the PACF plot finds correlation of residuals with the next lag value. An easier way includes using the auto-ARIMA function which determines the values of these parameters by searching through different combinations of the parameters and finding the least Akaike's Information Criterion (AIC). AIC is a widely used statistical model which quantifies using the goodness of fit. The lower the AIC, the better the model. The next step includes splitting the dataset into training and testing data. In this paper for the ARIMA model, 85% of the data is considered as training data where the model is trained and the remaining 15% is considered as testing data where the model is tested and evaluated. ARIMA model is imported from the statsmodels.tsa.arima.model. The model is built on the processed training data and the values are predicted using the ARIMA.fit().predict() function. The prices are forecasted using a 95% confidence interval using the .forecast() function.The performance of the model is evaluated on the tested and predicted data with the error metrics namely Root mean squared error, mean squared error and mean absolute error. The lower the error metrics, the better the performance of the model.

3.4. Working of the SARIMA Model

The SARIMA model is an extension of the ARIMA model with the seasonality component. This model cannot deal with large datasets since they cannot capture changing patterns in the large datasets. Being an extension of the ARIMA model, the SARIMA model involves both seasonal and non seasonal components to predict the stock prices using historical data provided. The seasonal component is represented by 'm' or 's' which is usually set as 12 for monthly data. The main advantage of using the SARIMA model over the ARIMA model is the involvement of the seasonal component which is useful in analyzing trends and patterns in the given stock data to correctly forecast the upcoming stock prices. The detailed working of the SARIMA model is described in the next section.

3.5. Building the SARIMA Model

The SARIMA model is an extension of the ARIMA model with the seasonality component. This model cannot deal with large datasets since they cannot capture changing patterns in the large datasets. Hence the data preprocessing required for this model includes converting the daily stock prices into a monthly stock price dataframe with an interval of 1 month. The next step includes seasonally decomposing the data to understand the different components like the trend in the data, the seasonality in the data, whether the data is stationary by observing the residuals. The next step is to find the values of the parameters p,d,q and P,D,Q,s where s is the seasonality in the data by the auto-ARIMA function. The next step is to split the data into training and testing dataset where 85% is the training dataset where the model is trained and 15% is the testing dataset where the model is tested and evaluated. The model is built using ARIMA.fit() function and the values are predicted using ARIMA.fit().predict() by mentioning the seasonal order of the model.[15] Further a graph can be plotted to check the accuracy of the model with the actual and predicted stock prices and the performance of the model is evaluated on the basis of the error metrics.

3.6. Working of the LSTM Model

LSTM is a type of Recurrent Neural Network (RNN) that needs a huge amount of historical data to be trained on. LSTM works by storing data in its memory cell and using it for further predictions. LSTM requires huge amounts of historical data to be trained which are normalized to ensure that features are evenly distributed over a range where X is all the features and Y is the target variable. Compared to ARIMA and SARIMA, LSTM model works better in finding the hidden trend patterns (non linear relationship) between the data provided. The detailed working of the LSTM model is described in the next section.

3.7. Building the LSTM Model

LSTM is a type of Recurrent Neural Network (RNN) that needs a huge amount of historical data to be trained on. LSTM works by storing data in its memory cell and using it for further predictions. [16] [17]Data preprocessing includes checking if there are any null columns present in the dataset followed by splitting the dataset into training and testing data by creating a matrix of all the columns in the dataset and providing our output data as the closing price of the particular bank stock. A condition called shuffle = False is set so that the data is not shuffled before splitting the dataset. The test size is set to 0.05 indicating that 95% of the data is used for training the model and 5% is used for testing and evaluating the model.

The next step includes scaling the data by the $MinMaxScaler()$ function from the sklearn.preprocessing library in the range of [0,1]. This is the most important part of data preprocessing since scaling the data improves the model performance and accuracy and improves the model's efficiency. Scaling the data is also a part of normalizing the data so that the data is used in the correct manner. This is done by importing the $MinMaxScaler()$ function from The first step in building the model is importing the necessary libraries which include the LSTM layer, the Dense Layer and the Dropout Layer from the library tensorflow.keras.layers for building the model. The next step includes importing the optimizer Adam from keras.optimizers. Adam is the most used optimizer in neural network modeling because it minimizes the loss function during the training and building of the model.

Next step includes importing the necessary error metrics which include root mean squared error, mean absolute error and mean squared error. Instead of using the traditional LSTM method of manually finding the parameters, we use an optimizer namely keras tuner which will help finding the best model after passing parameters through it. Defining the model includes using a function that takes hyperparameters as the input. The hyperparameters include integer hyperparameter tuning (hp.int()) for the number of units in LSTM layers, float hyperparameter tuning for the dropout rate. The dropout rate is a regularization technique in which some neurons from the LSTM layer are dropped out to prevent under or overfitting on the training data. Furthermore, a dense layer is added in the model with the help of Choice () function which allows you to pass possible values for hyperparameters and choose the best activation function for the layers for better model performance. It uses the dense_activation function for hyperparameter tuning. The possible values are ReLU (Rectified linear unit) and Sigmoid. [18]These functions allow the model to learn about

non linear relationships and complex patterns between different variables.

The next step involves initializing the keras tuner and searching for the best model which done by initializing a random search tuner that will take the model building function as the input and the objective is to reduce the mean squared error with a max trials of 4 and 2 models to be built and executed every trial.A directory would be used to save the results and a tuner.search() function with the parameters as the x_train and y_train sets with 10 epochs with a validation data : x_test and y_test will be used to run the hyperparameter search on the training data. The next step is to get the best model using tuner.get_best_models() and predict the future values using .predict() function. The last step is to calculate the error metrics namely Root mean squared error and mean absolute error.

4. Results

In this section the results of the stock price prediction of Indian bank stocks using ARIMA, SARIMA and LSTM is discussed.

4.1. Result analysis of ARIMA model

The ARIMA model performed well with the error metrics RMSE, MSE and MAE close to 0. For ICICI bank the auto ARIMA function provided the values for (p,d,q) to be (2,1,2) respectively. Upon fitting the model the Root mean squared error was 0.395, the mean squared error was 0.156 and the mean absolute error was 0.340 indicating a better model performance and prediction for the ARIMA model. For Kotak Mahindra Bank, the auto ARIMA function provided the values for (p,d,q) to be (3,1,3) respectively. Upon fitting the model the Root mean squared error was 0.262, the mean squared error was 0.0.68 and the mean absolute error was 0.214 indicating better model performance and pattern recognition by the ARIMA model. For the State Bank of India, the auto ARIMA function provided the values for (p,d,q) to be (0,1,1) respectively. Upon fitting the model, the root mean square error was 0.260, the mean square error was 0.677 and the mean absolute error was 0.194 indicating better model performance while predicting and forecasting the stock closing price on the data. For HDFC Bank, the auto ARIMA function provided the values for (p,d,q) to be (5,2,0) respectively. Upon fitting the model the Root mean squared error was 2.170, the mean squared error was 4.71 and the mean absolute error was 1.88 indicating a moderate model performance and stock closing price prediction for the ARIMA model. For Axis Bank, the auto ARIMA function provided the values for (p,d,q) to be (0,1,0) respectively. Upon fitting the model the Root mean squared error was 0.213, the mean squared error was 0.045 and the mean absolute error was 0.173 indicating better model performance and pattern recognition by the ARIMA model. For IndusInd Bank, the auto ARIMA function provided the values for (p,d,q) to be (2,1,4) respectively. Upon fitting the model, the root mean square error was 0.768, the mean square error was 0.591 and the mean absolute error was 0.598 indicating better model performance while predicting and forecasting the stock closing price on the data.

4.2. Result Analysis Of SARIMA Model

The SARIMA model performed well with the error metrics RMSE, MSE and MAE close to 0. For ICICI bank the auto ARIMA function provided the values for (p,d,q) and (P,D,Q,s) to be (1,0,2) and (1,1,1,12) respectively. The error metrics namely the Root mean squared error was 0.307 , the mean squared error was 0.094 and the mean absolute error was 0.261 which indicated better model performance and accuracy in plotting the predicted and forecasted values with the true values. For Kotak Mahindra Bank, the auto ARIMA function provided the values for (p,d,q) and (P,D,Q,s) to be (1,0,0) and (1,1,0,12) respectively. Upon fitting the model the error metrics namely root mean squared error turned out to be 0.154, the mean squared error metrics turned out to be 0.023 and the mean absolute error turned out to be 0.110 indicating better model performance on the bank closing price. For State Bank of India, the auto ARIMA function provided the values for (p,d,q) and (P,D,Q,s) to be (1,0,0) and (1,1,1,12) respectively. The error metrics namely the Root mean squared error was 0.332, the mean squared error was 0.110 and the mean absolute error was 0.247 which indicated better model performance and accuracy in plotting the predicted and forecasted values with the true values. For HDFC Bank, the auto ARIMA function provided the values for (p,d,q) and (P,D,Q,s) to be (2,0,1) and (2,1,1,12) respectively. The error metrics namely the Root mean squared error was 0.134, the mean squared error was

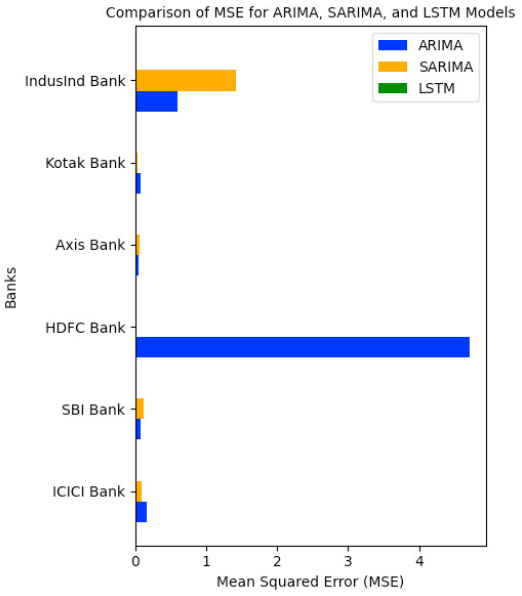


Fig. 2: MSE Values of the Models on Indian Banks

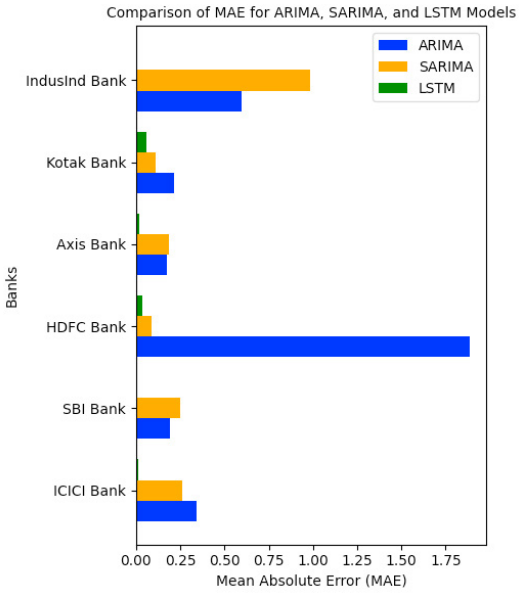


Fig. 3: MAE Values of the Models on Indian Banks

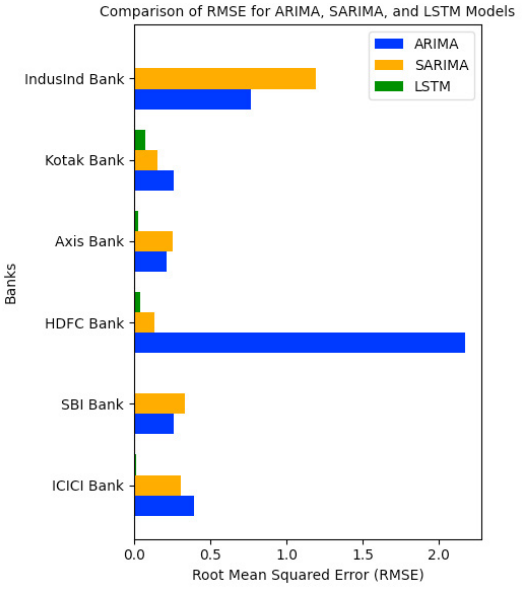


Fig. 4: RMSE Values of the Models on Indian Banks

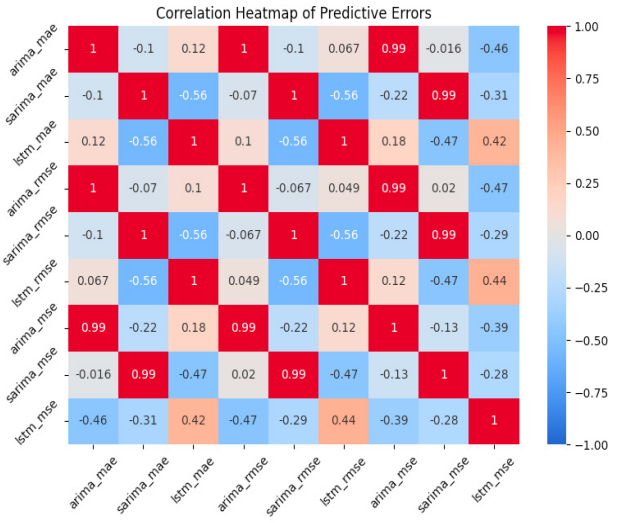


Fig. 5: Correlation Heatmap of Predictive Errors

0.018 and the mean absolute error was 0.088 which indicated better model performance and accuracy in plotting the predicted and forecasted values with the true values compared to the ARIMA model.For Axis Bank, the auto ARIMA function provided the values for (p,d,q) and (P,D,Q,s) to be (1,0,2) and (2,1,1,12) respectively. The error metrics namely the Root mean squared error was 0.256, the mean squared error was 0.065 and the mean absolute error was 0.185 which indicated better model performance and accuracy in plotting the predicted and forecasted values with the true values. For IndusInd Bank the auto ARIMA function provided the values for (p,d,q) and (P,D,Q,s) to be (0,1,0) and (2,1,0,12) respectively. The error metrics namely the Root mean squared error was 1.192, the mean squared error was 1.420 and the mean absolute error was 0.982 which indicated better model performance and accuracy in plotting the predicted and forecasted values with the true values.

4.3. Result Analysis of LSTM Model

The LSTM model performed well after 4 trials of testing and validating the data using the kerastuner hyperparameter optimizer. For ICICI bank the best model turned out to be with the mean squared error of 2.29e-05, a root mean squared error of 0.017, a mean absolute error of 0.012 and a mean absolute percentage error of 1.68%. Considering lower error metric values the model is performed well on the ICICI stock price data and is able to predict the future prices. For Kotak Mahindra Bank, the best model turned out to be with a mean squared error to be 0.0005 with a root mean squared error of 0.077, mean absolute error of 0.061 and a mean absolute percentage error of 7.07%. Considering the lower error metrics namely RMSE, MSE and MAE but a high MAPE it can be considered the model is able to capture the complex patterns in the dataset but while predicting and forecasting the values there is an error of 7% in the stock closing price prediction data. For State Bank of India the best model turned out to be with the mse of 0.00019 with root mean squared error to be 0.010, mean absolute error to be 0.008 and mean absolute percentage error to be 1.64% indicating that the model was trained well on the training dataset and was able to predict and forecast the values of the stock closing price. For HDFC Bank the best model turned out to be with the mse of 8.80e-05 with root mean squared error to be 0.038, mean absolute error to be 0.032 and mean absolute percentage error to be 4.05% indicating that the model was trained well on the training dataset and was able to moderately predict and forecast the values of the stock closing price. For Axis Bank the best model turned out to be with the mse of 0.0008 with root mean squared error to be 0.025, mean absolute error to be 0.020 and mean absolute percentage error to be 2.92% indicating that the model was trained well on the training dataset and was able to predict and forecast the values of the stock closing price.For IndusInd Bank the best model turned out to be with the mse of 0.0006 with root mean squared error to be 0.0115, mean absolute error to be 0.009 and mean absolute percentage error to be 2.80% indicating that the model was trained well on the training dataset and was not correctly able to predict and forecast the values of the stock closing price.
Following observations can be made from these results

- According to figure 2, LSTM model performs well compared to ARIMA and SARIMA in terms of lower RMSE on all banks for predicting the closing stock price.
- According to figure 3 and 4 , the error metrics state that the ARIMA model has failed to capture the complex patterns in the model leading to huge errors.

Figure 5 shows the heatmap correlation of all error metrics namely Root Mean Squared Error, Mean absolute Error and Mean Squared Error for all the models used namely, ARIMA,SARIMA and LSTM. Heatmap is a part of exploratory data analysis and is used to find relationship between different variables or how correlated they are to each other. Key observations from figure 4 is as follows:

- The first key observation from the heatmap is the diagonal elements (1) which show that a particular error metric for a particular model are perfectly correlated with other.
- The next key observation is for the respective models, ARIMA and SARIMA, their error metrics, mean absolute error along with mean square error, and root mean square error along with mean square error and vice versa, the correlation metric is 0.99 which shows that each error metric is closely related to each other and they are capable of finding the same predictive error for a particular model.
- The next key observation is for the LSTM Model along with its predictive error metrics namely MAE along with RMSE and RMSE along with MSE respectively show a correlation error metric of 0.44 which states a positive weak correlation and that LSTM follows a different methodology for measuring errors in comparision to ARIMA and SARIMA models. For the same model, the error metric MAE along with MSE shows a correlation of 0.42 which states that when these metrics are used for the LSTM model, the model behaves differently when capturing errors as seen with the error metrics, RMSE and MSE.
- The next key observation is between using the same error metric, MAE but for 2 different models, SARIMA and ARIMA and it is seen from the heatmap that the correlation is -0.1 indicating that the models capture trends in the data differently as discussed earlier in this paper. The same negative correlation is visible for the error metric RMSE for the model SARIMA with ARIMA (-0.067) and SARIMA with LSTM (-0.56) respectively.

From the above observations, it can therefore be concluded that the ARIMA Model works in a different manner as compared to SARIMA and LSTM. The negative correlation observed between SARIMA and LSTM tells us that they work in opposite direction i.e if one models performs well, the other might not do so and vice versa. However, SARIMA and LSTM have proven to be the best chosen prediction and validation models for Indian stock price prediction.

Conclusion

Stock price prediction plays an important role in the finance industry. Therefore it is important to use the correct model for prediction. We can conclude from this research that SARIMA and LSTM are proven to be the best models for stock price prediction of the Indian bank stocks due to high volatility, seasonality and complex patterns in the data. ARIMA model is suitable for univariate time series forecasting and prediction since it captures linear related data easily whereas the SARIMA model being an extended version of ARIMA model is better at capturing seasonality and trends in the data with lower error metrics compared to ARIMA model. LSTM model has proven to be the best model for this research because of its ability to read complex patterns in the data and then use these patterns to give accurate predictions for stock prices. It can now be concluded that SARIMA can be used for data where there is strong seasonality and LSTM can be used to capture complex non linear relationships between different data. Future work could include creating a hybrid model consisting of SARIMA and LSTM so that they can together capture seasonality as well as the complex patterns and improve accuracy and precision in predicting the prices of the stocks. To conclude, LSTM provides the highest accuracy followed by SARIMA and then ARIMA model for this research.

References

[1] J. Patel, S. Shah, P. Thakkar, K. Kotecha, Predicting stock market index using fusion of machine learning techniques, Vol. 42, Elsevier, 2015.
[2] D. M. Nelson, A. C. Pereira, R. A. d. Oliveria, Stock market's price movement prediction with lstm neural networks, IEEE, 2017.
[3] L. Pant, S. Ratogi, Stock market prediction using machine learning techniques, Vol. 42, Foundation of cumputer science (FCS),USA, 2019.
[4] H. Leung, P. Mak, Stock market prediction using deep learning, Elsevier, 2020.
[5] Y. Shen, H. Jiang, An ensemble approach for stock market prediction using machine learning, Elsevier, 2018.
[6] J. Jones, C. Brown, Application of holt-winters and sarima models for stock price prediction, Elsevier, 2019.
[7] E. Zhang, L. Zhao, Predicting stock prices using a combined arima and neural network model, Elsevier, 2019.
[8] U. M. Sirisha, M. C. Belavagi, G. Attigeri, Profit prediction using arima, sarima and lstm models in time series forecasting: A comparison, Vol. 10, 2022.
[9] A. Mahadik, D. Vaghela, A. Mhaisgawali, Stock Price Prediction using LSTM and ARIMA, 2021.
[10] O. Deorukhkar, S. Lokhande, V. Nayak, A. Chougule, Stock price prediction using combination of lstm neural networks, arima and sentiment analysis, 2019.
[11] A. Dutta, J. Sen, S. Mehtab, Stock price prediction using machine learning and lstm based deep learning models, springer, 2021.
[12] K. Hoque, H. Aljamaan, Impact of hyperparameter tuning on machine learning model in stock price forecasting, IEEE, 2021.
[13] A. Chatterjee, H. Bhowmick, J. Sen, Stock price prediction using time series, econometric, machine learning and deep learning models, in: 2021 IEEE mysore sub section international conference(MysuruCon), 2021.
[14] J. Fattah, L. Ezzine, Z. Aman, H. Moussami, Forecasting of demand using arima model, in: International journal of engineering business management, 2018.
[15] S. Sikhakolli, C. Manideep, R. Rajyalakshmi, A. Deshpande, Lstm based stock price prediction on daily charts, in: 2023 4th international conference on intelligent technologies (CONIT), 2024.
[16] M. Maswood, A. Alharbi, Deep learning based stock price prediction using lstm and bi directional lstm model, 2020.
[17] A. Gupta, D. Kapil, N. Jain, Abhishek, H. singh, Using neural netwrok for financial forecasting, in: 5th international conference for emerging technology(INCET), 2024.
[18] R. Vinayakumar, E. Gopalakrishnan, V. Menon, Stock price prediction using lstm, rnn and cnn sliding window model, 2017.